

9. Refer to Exercise 4 (Section 15.2). The data in this table are from a paired-difference experiment with $n = 7$ pairs of observations.

Population	Pair						
	1	2	3	4	5	6	7
1	8.9	8.1	9.3	7.7	10.4	8.3	7.4
2	8.8	7.4	9.0	7.8	9.9	8.1	6.9

- a. Use Wilcoxon's signed-rank test to determine whether there is a significant difference between the two populations.
- b. Compare the results of part a with the result you got in Exercise 4 (Section 15.2). Are they the same? Explain.

a. H_0 : 2母體無顯著差異
 H_a : 2母體有顯著差異

$$T^+ = 26.5$$

$$T^- = 1.5$$

$$T = \min(T^+, T^-) = 1.5$$

$$n=7, \alpha=0.05 \Rightarrow 2$$

$$1.5 < 2 \rightarrow \text{reject } H_0$$

H_0 : 兩者無顯著差異

H_a : 兩者有顯著差異

$$T^+ = 6.5 \quad T^- = 29.5$$

$$T = \min(T^+, T^-) = 6.5$$

$$n=8, \alpha=0.05, \text{臨界值} = 4$$

$$6.5 > 4 \rightarrow \text{Non-reject}$$

11. **Machine Breakdowns** The number of machine breakdowns per month was recorded for 9 months on two identical machines, A and B, used to make wire rope:

Month	A	B		
1	3	7	-4	-6
2	14	12	-2	6.5
3	7	9	-2	-3.5
4	10	15	-5	-1
5	9	12	-3	-5
6	6	6	0	0
7	13	12	1	1.5
8	6	5	1	1.5
9	7	13	-6	-8

- a. Do the data provide sufficient evidence to indicate a difference in the monthly breakdown rates for the two machines? Test by using a value of α near .05.
- b. Can you think of a reason the breakdown rates for the two machines might vary from month to month?

15. Images and Word Recall A psychology class performed an experiment to determine whether a recall score in which instructions to form images of 25 words were given differs from an initial recall score for which no imagery instructions were given. Twenty students participated in the experiment with the results listed in the table.

Student	With Imagery	Without Imagery	Student	With Imagery	Without Imagery
1	20	5	11	17	8
2	24	9	12	20	16
3	20	5	13	20	10
4	18	9	14	16	12
5	22	6	15	24	7
6	19	11	16	22	9
7	20	8	17	25	21
8	19	11	18	21	14
9	17	7	19	19	12
10	21	9	20	23	13

- What three testing procedures can be used to test for differences in the distribution of recall scores with and without imagery? What assumptions are required for the parametric procedure? Do these data satisfy these assumptions?
- Use both the sign test and the Wilcoxon signed-rank test to test for differences in the distributions of recall scores under these two conditions.
- Compare the results of the tests in part b. Are the conclusions the same? If not, why not?

Sign Test

H_0 : 無顯著差距

H_a : 有顯著差距

$+$: 20, $-$: 0, $n=20$

$$p\text{-value} = 2(P \times 50)$$

$$= 2 \times 0.5^{20} < 0.05$$

$\rightarrow$ reject H_0

Wilcoxon Signed

$$T^+ = 210, T^- = 0 \Rightarrow T = 0$$

$$T = 0 < 32 \rightarrow \text{reject } H_0$$

Applying the Basics

6. Legos® The time required for kindergarten children to assemble a specific Lego creation was measured for children who had been instructed for four different lengths of time. Five children were randomly assigned to each instructional group. The length of time (in minutes) to assemble the Lego creation was recorded for each child in the experiment.

Training Period (hours)				
.5	1.0	1.5	2.0	
8	15	9	16	4
14	20	7	10	6
9	16	5	3	7
12	19	8	16	8
10	18	9	16	6

$$H = \frac{12}{20 \cdot 21} \cdot \left(\frac{1396}{5} + \frac{3600}{5} + \frac{1600}{5} + \frac{376}{5} \right) - 3 \cdot 21$$

$$\approx 12.25$$

$$\chi^2_{3,0.01} = 11.34 \rightarrow \text{reject } H_0$$

Use the Kruskal-Wallis H -Test to determine whether there is a difference in the distribution of times for the four different lengths of instructional time. Use $\alpha = .01$.

8. Heart Rate and Exercise In Exercise 18 (Section 11.3), we presented data on the heart rates for samples of 10 men randomly selected from each of four age groups. Each man walked a treadmill at a fixed grade for a period of 12 minutes, and the increase in heart rate (the difference before and after exercise) was recorded (in beats per minute). The data are shown in the table.

	10-19	20-39	40-59	60-69
29	24	37	28	
33	27	25	29	
26	33	22	34	
27	31	33	36	
39	21	28	21	
35	28	26	20	
33	24	30	25	
29	34	34	24	
36	21	27	33	
22	32	33	32	
Total	309	275	295	282

- a. Do the data present sufficient evidence to indicate differences in location for at least two of the four age groups? Test using the Kruskal-Wallis H -test with $\alpha = .01$.
- b. Find the approximate p -value for the test in part a.
- c. Since the F -test in Chapter 11 and the H -test are both tests to detect differences in location of the four heart-rate populations, how do the test results compare? If the ANOVA F -test produced the test statistic $F = 0.87$ with p -value = 0.468, compare this p -value with the p -value in part b and explain the implications of the comparison.

H_0 : 四種組別分佈沒有差異

H_a : 至少一組不同

$$H = \frac{12}{40(41)} \left(\frac{247.5^2}{10} + \frac{168^2}{10} + \frac{216.5^2}{10} + \frac{188^2}{10} \right) - 3 \times 41$$

$$= 2.632$$

$$\chi^2_{3,0.01} = 11.345 \quad \chi^2 = 2.632 < 11.345 \rightarrow \text{Non reject}$$

10. Good Tasting Medicine A voluntary sample of healthy children was used in a study to assess their reactions to the taste of four antibiotics.⁷

The children's response was measured on a visual scale using faces, from sad (low score) to happy (high score). The minimum score was 0 and the maximum was 10. For the accompanying data (based on the results of the study), each of five children was asked to taste each of four antibiotics and rate them using the visual (faces) scale from 0 to 10.

Child	Antibiotic			
	1	2	3	4
1	4.8	2.2	1	6.8
2	8.1	9.2	3	6.6
3	5.0	2.6	1	3.6
4	7.9	9.4	4	5.3
5	3.9	7.4	4	2.1

- a. What design is used in collecting these data?
- b. The normal probability plot of the residuals as well as a plot of residuals versus antibiotics are shown below. Do the usual analysis of variance assumptions appear to be satisfied?

a. randomized block design

H_0 : 四種分佈完全相同

H_a : 不完全相同

$$F_r = \frac{12}{5 \cdot 4 \cdot 5} (12^2 + 13^2 + 10^2 + 14^2) - 3 \cdot 5 \cdot 5$$

$$= 1.56$$

$$1.56 < \chi^2_{3,0.05} = 7.8$$

$\rightarrow$ Non reject